

JIYOUNG LEE

🔗 <https://github.com/jiyounglee-0523> 🌐 <https://jiyounglee-0523.github.io> ✉ jiyounglee0523@kaist.ac.kr

RESEARCH INTEREST

MAIN INTEREST: Building Benchmark Datasets for Evaluating LLM Safety and Alignment

KEYWORDS: Dataset Construction, LLM Evaluation

I have extensive experience in building benchmark datasets for evaluating large language models, particularly with respect to safety and alignment, including the design of scenario-driven tasks and rigorous evaluation protocols. I have led large-scale human studies, designing and conducting surveys with over 6,000 participants to collect high-quality human judgments that ground and validate model behavior. In parallel, I have collaborated closely with domain experts across multiple fields to refine task definitions, label taxonomies, and evaluation criteria, ensuring that benchmarks accurately reflect real-world risks and requirements. I have also constructed in-house training and evaluation datasets for LLMs, from data specification and collection to cleaning, labeling, and split design, enabling controlled experimentation and domain-specific fine-tuning for both research and production use cases.

EDUCATION

Ph.D in Artificial Intelligence

Sep 2022 - Present

Korea Advanced Institute of Science and Technology (KAIST), Daejeon, South Korea

Advisor: Prof. Edward Choi

M.Sc. in Artificial Intelligence

Sep 2020 - Aug 2022

Korea Advanced Institute of Science and Technology (KAIST), Daejeon, South Korea

Advisor: Prof. Edward Choi

B.Sc. in Statistics

March 2016 - Aug 2020

Sookmyung Women's University, Seoul, South Korea

GPA: 4.08 / 4.3 (Ranked 1st in the Department of Statistics)

WORK EXPERIENCE

AI Engineer Intern

Sep 2025 - Present

LG AI Research - EXAONE Team

Machine Translation Research Intern

Feb 2022 - Aug 2022

Naver Corporation - Papago Team

Specializing Multi-domain NMT via Penalizing Low Mutual Information (EMNLP 2022)

PROFESSIONAL SERVICE

Advisory Committee Member

Sep 2023 - Feb 2024

SelectStar - LLM Dataset Construction

KorNAT: LLM Alignment Benchmark for Korean Social Values and Common Knowledge (ACL 2024)

PUBLICATIONS

Conference Proceedings

- **Jiyoung Lee**, Seungho Kim, Jieun Han, Jun-Min Lee, Kitaek Kim, Alice Oh, and Edward Choi. Trans-EnV: A Framework for Evaluating the Linguistic Robustness of LLMs Against English Varieties. In *Proc. of Neural Information Processing Systems (NeurIPS) 2025 Datasets and Benchmarks*
- Soyoung Yang, Hojun Cho, **Jiyoung Lee**, Sohee Yoon, Edward Choi, Jaegul Choo, and Won Ik Cho. Single Ground Truth Is Not Enough: Add Linguistic Variability to Aspect-based Sentiment Analysis Evaluation. In *Proc. of Conference on Annual Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics (NAACL) 2025*

- **Jiyoung Lee**, Minwoo Kim, Seungho Kim, Junghwan Kim, Seunghyun Won, Hwaran Lee, and Edward Choi. KorNAT: LLM Alignment Benchmark for Korean Social Values and Common Knowledge. In *Findings in Association for Computational Linguistics (ACL) 2024*
- **Jiyoung Lee**, Seungho Kim, Seunghyun Won, Joonseok Lee, Marzyeh Ghassemi, James Thorne, Jaeseok Choi, O-Kil Kwon, and Edward Choi. VisAlign: Dataset for Measuring the Degree of Alignment between AI and Humans in Visual Perception. In *Proc. of Neural Information Processing Systems (NeurIPS) 2023 Datasets and Benchmarks*
- Woncheol Shin, Gyubok Lee, **Jiyoung Lee**, Eunyi Lyoo, Joonseok Lee, and Edward Choi. Translation-equivariant Image Quantizer for Bi-directional Image-Text Generation. In *Proc. of International Conference on Acoustics, Speech, and Signal Processing (ICASSP) 2023, (Oral Presentation)*
- **Jiyoung Lee**, Hantae Kim, Hyunchang Cho, Edward Choi, and Cheonbok Park. Specializing Multi-domain NMT via Penalizing Low Mutual Information. In *Proc. of Conference on Empirical Methods in Natural Language Processing (EMNLP) 2022*
- Kyunghoon Hur*, **Jiyoung Lee***, Jungwoo Oh, Wesley Price, Young-Hak Kim, and Edward Choi. Unifying Heterogenous Electronic Health Records Systems via Text-Based Code Embedding. In *Proc. of Conference on Health, Inference, and Learning (CHIL) 2022*

Workshops and Preprints

- **Jiyoung Lee**, Seungho Kim, Jieun Han, Jun-Min Lee, Kitaek Kim, Alice Oh, Edward Choi. Trans-EnV: A Framework for Evaluating the Linguistic Robustness of LLMs Against English Varieties. *arXiv:2505.20875 2025*
- Radhika Dua, **Jiyoung Lee**, Joon-myung Kwon, and Edward Choi. Automatic Detection of Noisy Electrocardiogram Signals without Explicit Noise Labels. In *International Workshop on Pattern Recognition in Healthcare Analytics (PRHA) 2022*
- **Jiyoung Lee**, Wonjae Kim, Daehoon Gwak, and Edward Choi. Conditional Generation of Periodic Signals with Fourier-Based Decoder. In *Deep Generative Models and Downstream Applications Workshop at NeurIPS 2021*

TECHNICAL SKILLS

Program Language	Python
Packages	Pytorch, HuggingFace, LangChain, LangGraph, vLLM
Language	Korean (Native), English (Fluent)

EXPERIENCES

Lecture	KoSAIM Summer School - Chest X-ray Classification using CNN Hands-on session (Aug 2025) KoSAIM Summer School - Numpy and Pytorch Intro Hands-on session (Aug 2020)
Teaching Assistant	AI504: Programming for AI, KAIST, 2024, 2023, 2021, 2020 Fall AI612: Machine Learning for Healthcare, KAIST, 2023, 2021 Spring
Academic Service	ARR May Reviewer NeurIPS 2025 Reviewer ACL 2025 Reviewer ACCV 2022 Reviewer

AWARDS & SCHOLARSHIPS

NeurIPS 2023 Travel Award
Google Conference Scholarship 2022
National Science & Technology Scholarship (2018 - 2020)

REFERENCES

-
- Edward Choi, Associate Professor at Kim Jaechul Graduate School of AI, KAIST., edwardchoi@kaist.ac.kr
 - Hwaran Lee, Assistant Professor in the Department of Artificial Intelligence and Computer Science Engineering, Sogang University., hwaranlee@sogang.ac.kr

- Sungjin Park, PhD at KAIST, Past Microsoft Asia Internship., zxznm@kaist.ac.kr
- Cheonbok Park, Researcher at Foundation Research, Hyperscale AI, NAVER Corp., cbok@navercorp.com